SULTAN MAHMUD

Backend Engineer

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PROFESSIONAL SUMMARY

Backend Engineer specializing in **distributed systems** and **Kubernetes infrastructure**. Built **production-grade** portfolio projects with **automated deployment**: **ML-enhanced K8s autoscaler** (**60% cost reduction**), **polyglot microservices** (**sub-1ms latency**), managing **11+ EC2 instances** across **multi-AZ AWS infrastructure**. Deep expertise in **AWS**, **Python**, **Go**, **observability stacks** (**OpenTelemetry**, **Tempo**, **Grafana**). Seeking **full-time platform engineering** role.

EXPERIENCE

BACKEND ENGINEER & PRODUCT BUILDER | *Remote* *August 2024 - Present*

Building production systems to demonstrate platform engineering capabilities while actively seeking full-time opportunities

- **Architected and deployed 5 production-grade applications with complete CI/CD pipelines** across **e-commerce**, **fintech**, and **SaaS** domains, managing **11+ EC2 instances** with automated **monitoring and alerting** through **multi-AZ AWS infrastructure**
- Built **ElastiKube**: **ML-enhanced Kubernetes autoscaler** with **4-layer intelligent scaling** (**time-aware**, **flash sale detection**, **Prophet forecasting**), achieving **60% cost reduction** through **event-driven Lambda architecture** and **multi-AZ high availability**
- Engineered **polyglot URL shortener**: **Go** `redirect_service` (**sub-1ms latency**), **Python** `create_service`, with **comprehensive observability stack** (**OpenTelemetry**, **Tempo**, **Grafana**, **Loki**) and **circuit breaker fault tolerance**
- Deployed **production infrastructure** using **Pulumi IaC** and **Ansible** across **multi-AZ VPC**, implementing **automated failover**, **distributed tracing**, and **zero-downtime CI/CD pipelines**

BACKEND DEVELOPER | *Cooking Station, Dhaka* *June 2024 - August 2024*

- **Streamlined operations** for 200+ users by designing **role-based admin dashboard** with **real-time meal analytics**, reducing **manual subscription management overhead**
- **Eliminated 40% of manual effort** in account management through **automated balance updates** and **error-prevention systems**
- Built **production-ready meal scheduling system** using **cron jobs** with **configurable time boundaries** and **daily user listing functionality**

NOTABLE PROJECTS

ELASTIKUBE: PRODUCTION K3S AUTOSCALER FOR AWS | Demo

Challenge: Build production-grade intelligent autoscaler with ML-based predictive scaling for K3s clusters on AWS

Impact: Event-driven Lambda architecture with 4-layer intelligent scaling achieving 60% cost reduction ($200→$111-$149/month), multi-AZ high availability, and proactive pre-scaling before traffic spikes

Technical Achievements:

- Architected **4-layer intelligent autoscaling**: (1) **Data Collection**, (2) **Time-Aware Scaling** (peak/off-peak: 85%/60% vs 60%/40%), (3) **Flash Sale Detection** (>30% spike in 2min), (4) **Prophet ML forecasting CPU 15min ahead**
- **Built ML training pipeline** with **Kubernetes CronJob** for weekly automated retraining, **feature engineering** (cyclical encoding, lag features), **cross-validation**, and **backtesting**
- **Implemented distributed locking** via **DynamoDB conditional writes**, **Write-Ahead Log (WAL)** for crash recovery, and **EventBridge orchestration** for fault-tolerant scaling
- Engineered **multi-AZ high availability** with round-robin distribution across 3 AZs, **LIFO scale-down** with **kubectl drain** via **SSM**, and single **NAT Gateway** optimization
- Deployed comprehensive **observability**: 17 **CloudWatch alarms** (CRITICAL/WARNING), **Prometheus graceful degradation**, fixed **LogGroups**, **DLQ monitoring**, and **SNS alerting**
- Built **spot instance support** with automatic **On-Demand fallback** handling 3 **AWS capacity exceptions** for **70% cost savings**

Key Innovation: Hybrid reactive+proactive scaling combining **time-aware thresholds**, **flash sale detection**, and **ML forecasting** for **SLA-driven pre-scaling**

ULTRA-FAST URL SHORTENER MICROSERVICE

Challenge: Design distributed URL shortener with independent service scaling and sub-10ms redirects

Impact: Achieved 83% latency reduction (10ms → sub-1ms) by porting redirect service to Go, deployed comprehensive observability stack with Tempo tracing and Loki log aggregation

Technical Achievements:

- **Architected** polyglot microservices: **Go redirect_service** (**95% read traffic, sub-1ms**), **Python** `create_service` (write-heavy), `Celery worker_service` with independent **scaling**.
- **Implemented** high-performance Go redirect service using **Chi router**, internal/clean architecture, **circuit breaker pattern** (Closed → Open → Half-Open), and **cache-aside pattern** with **30-minute Redis TTL**.
- **Engineered complete observability**: **OpenTelemetry instrumentation** with **OTLP collector**, **Tempo** distributed tracing, **Loki** log aggregation via **Promtail**, and **Grafana** dashboards for end-to-end service visibility.
- **Built repository pattern** with abstract base classes for **PostgreSQL**/**MongoDB**, **optimized key allocation** using **SELECT FOR UPDATE SKIP LOCKED** for **race-free operations**, and parameterized **bulk inserts** (**100K+ keys**).
- **Implemented intelligent key pre-population** using **Celery workers** maintaining a pool of unused **short URL keys** for **instant URL creation** without database latency.
- **Engineered production resilience**: **PgBouncer connection pooling** achieving **53% overhead reduction** with **exponential backoff retries** and database **timeout protection**.
- **Built comprehensive testing infrastructure**: **multi-database mocking** (SQLite, mongomock, fakeredis), **async pytest** framework, **httpx API client testing**, and isolated test environments.
- **Deployed** on **K3s/AWS** via **Pulumi IaC**, **Ansible** configuration, and **GitHub Actions CI/CD** with automated infrastructure provisioning.

Key Innovation: Polyglot architecture demonstrating pragmatic technology selection—**Go** for performance-critical paths (**sub-1ms redirects**), **Python** for business logic, enabling **independent service scaling** without rewrites.

TECHNICAL SKILLS

- **LANGUAGE & FRAMEWORKS: Python,** Golang, JavaScript, **FastAPI**, Django, DjangoREST
- **CLOUD & INFRASTRUCTURE: AWS**(EC2, S3, VPC, Lambda, EventBridge, CloudWatch, SNS, SQS), **Docker**, Kubernetes(K3s), Pulumi, Ansible
- **DEVOPS & CI/CD**: GitHub Actions, Docker Compose, Nginx, Load Balancing
- **ARCHITECTURE & DESIGN:** Microservices, Distributed Systems, High Availability, Fault Tolerance, Load Balancing, Caching Strategies, Database Optimization
- **DATABASE & CACHING: PostgreSQL**, **Redis**, MongoDB, SQLAlchemy, PgBouncer (Connection Pooling), Alembic (Migrations)
- **API DEVELOPMENT:** Swagger/OpenAPI, RESTful API, Authentication (JWT), Rate Limiting, CORS
- **MESSAGE QUEUING & ASYNC: Celery(Worker, Beat, Flower)**, **Redis**, RabbitMQ, Async/Await
- **OBSERVABILITY & MONITORING:** OpenTelemetry (Distributed Tracing), Prometheus, Grafana, Tempo

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EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING

Daffodil International University, Dhaka *September 2017 - September 2023*

EXTRACURRICULAR ACTIVITIES & COMMUNITY ENGAGEMENT

- **Quora Bengali:** Maintain A Dedicated Blog On **Quora** Offering Insights And Solutions To Inquiries Spanning Technology, Culinary Arts, And Health In Bengali. Accumulated **Readership** Nearing **200,000** With Nearly **200 Followers**.
- **Medium Technical Blog:** Published 2 in-depth technical articles (1,000+ words each) discussing distributed systems, backend architecture & engineering decisions and trade-offs
- **YouTube Channel:** Founded And Managed The Youtube Channel "**I.T. Darshonik**" Producing Over **40 Instructional Videos** In Bengali, Aimed At Bridging The Language Gap In Tech Education.

AWARDS & ACHIEVEMENTS

- **DIU Take-Off Programming Contest, Spring 2018:** Ranked **6th/300+ participants** (Top 2%)
- **Competitive Programming:** Solved 500+ problems across BeeCrowd, LightOJ, HackerRank, LeetCode.
- **Proficiency:** Dynamic Programming, Graph Theory, Greedy Algorithms, Binary Search